

karAQI — 72-Hour Air Quality Index Predictor

End-to-End Machine Learning and Automated CI/CD for Karak, Pakistan

Developed by Faisal Iqbal (NUST / 10Pearls Shine Internship 2026)

Karak's Air Quality Crisis

No Ground Monitors for a Vulnerable Region

Karak operates in an absolute data void—zero local air quality monitoring stations. Commercial APIs are either unavailable or over 6+ hours stale. This leaves the population critically exposed to **unmonitored** environmental hazards:



Pre-monsoon dust
storms (March–June)



Agricultural biomass
burning



Severe seasonal PM2.5
temperature inversions

Leveraging Open-Meteo and Satellite Reanalysis

Bypassing Physical Sensors with Modeled Reanalysis

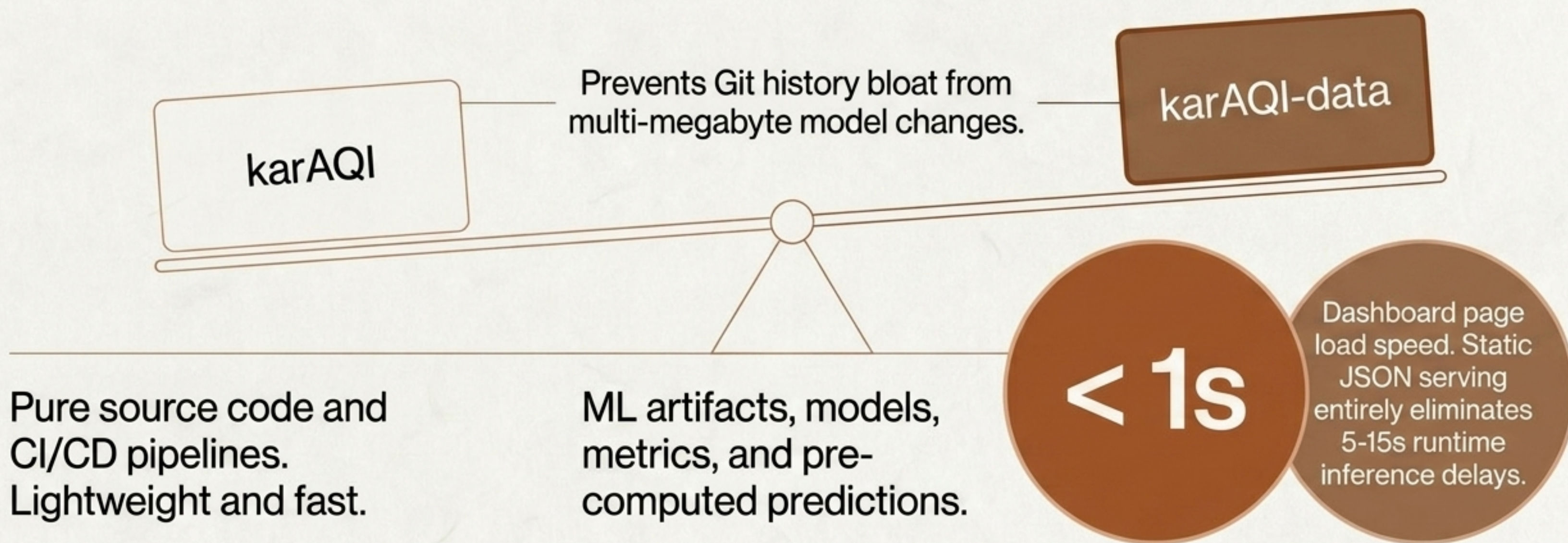
By utilizing Open-Meteo's CAMS (Copernicus Atmosphere Monitoring Service) reanalysis, karAQI models pollutant concentrations at an hourly resolution. This satellite-derived approach entirely removes the dependency on physical ground stations.

$$r = 0.68$$

Open-Meteo's correlation with the best available IQAir readings after timezone normalization.

The Two-Repository Design Pattern

Clean Separation of Code and Heavy ML Data



Analytical DuckDB Engine

Analytical Feature Storage & Embargo Ingestion

Utilizes DuckDB as an analytical file-based database for incremental feature backfilling.

To strictly prevent time-series data leakage, the pipeline enforces a 72-hour training embargo window.

35,400

Rows of engineered hourly features spanning 33 variables, including lag, rolling, and cyclical encodings.

Championing XGBoost in Production

Comparative Performance and Infrastructure Constraints

XGBoost (Champion)

Selected for hourly predictions;
captures non-linear
interactions natively.

Ridge Regression

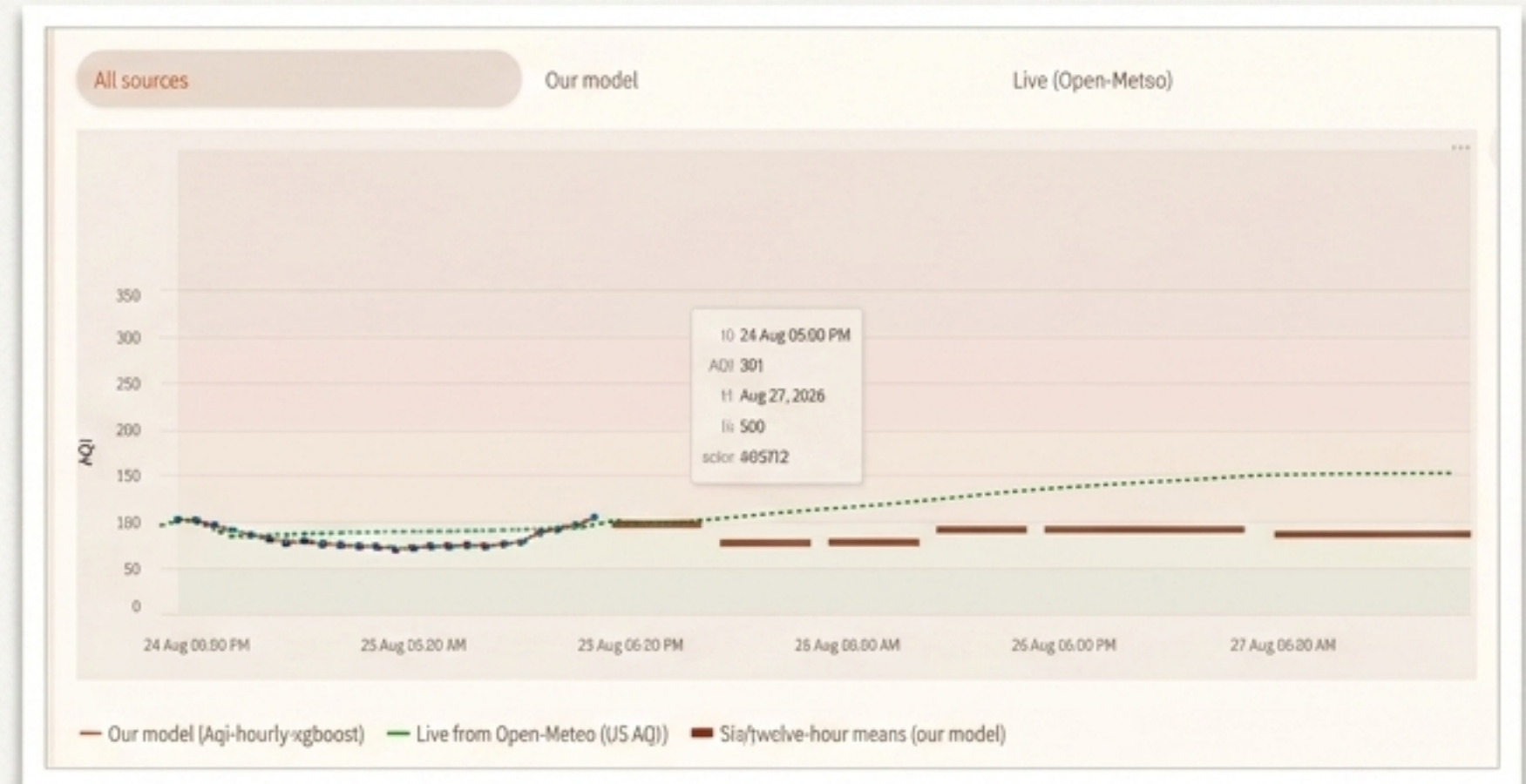
Selected for stable daily
baseline trends.

Random Forest

Pruned. 4.25 GB file exceeds
strict GitHub 100MB limits.

LSTM

Pruned. 1+ hour CPU training
incompatible with daily CI/CD
runners.



Beating Persistence by 38%

Expanding-Fold Rolling-Origin Assessment

Rigorous evaluation conducted across 3 expanding folds and 168-hour test windows to mirror true production conditions.

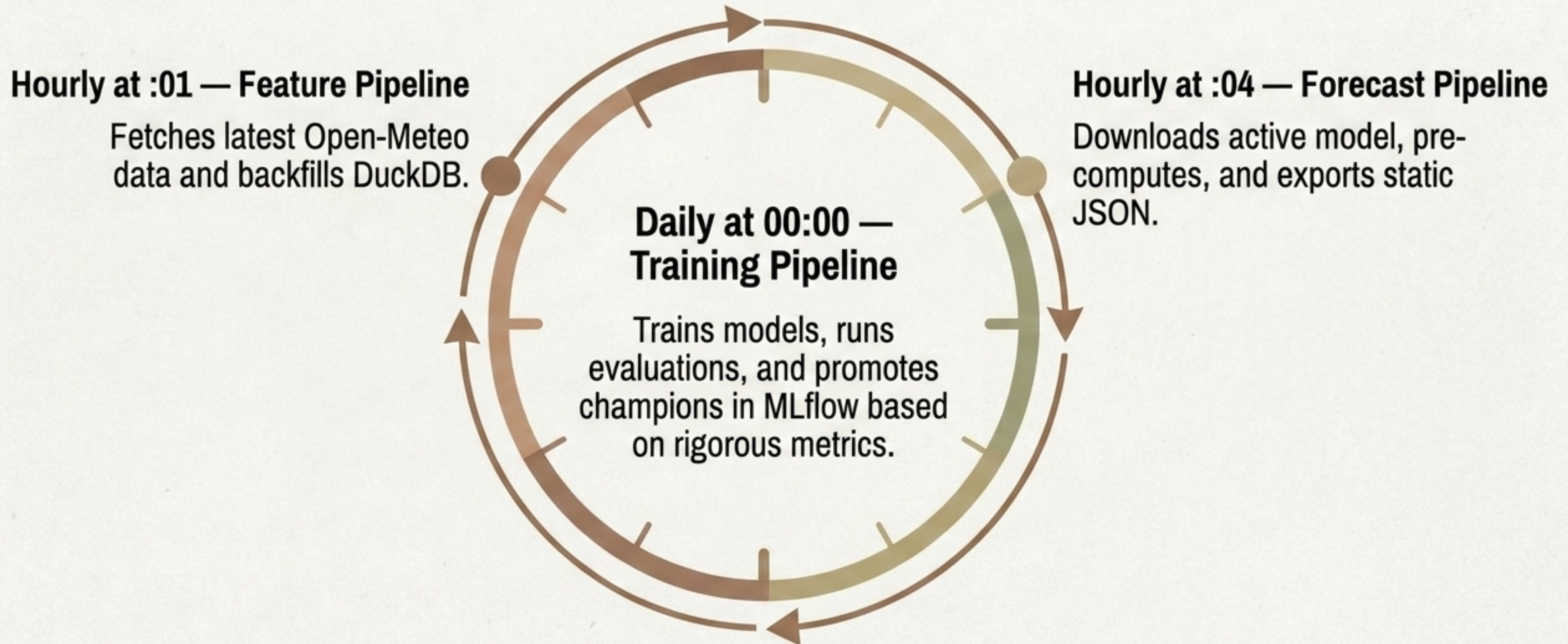
Metric	1-24h Hourly Points
RMSE	8.89
MAE	6.43
R ²	0.845
Category Accuracy	89.3%

38%

Improvement of our champion XGBoost model over the naive persistence baseline on hourly points.

GitHub Actions CI/CD Pipeline

The Hands-Free 24/7 Forecasting Clockwork

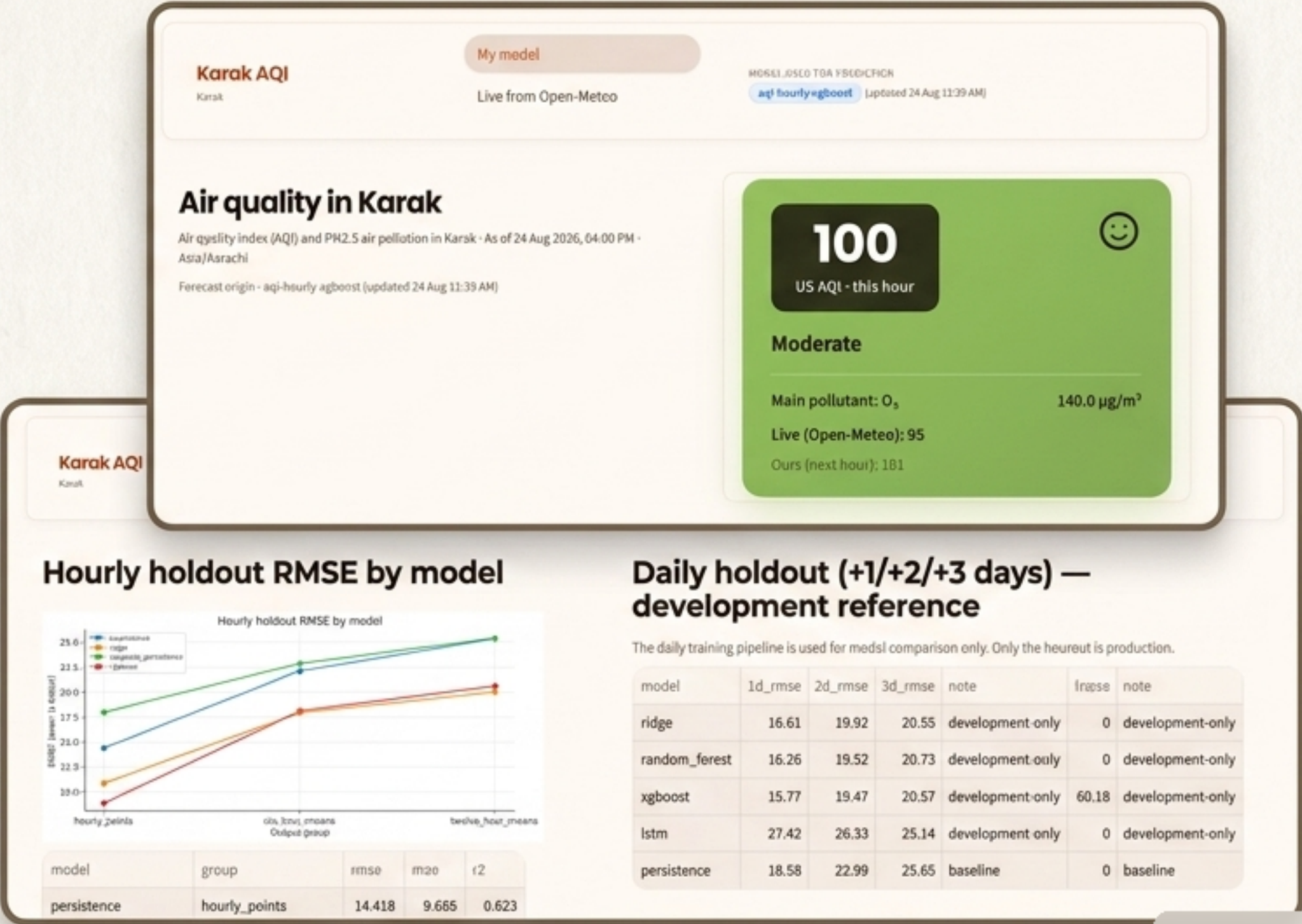


Streamlit-Powered Analytics

An Elegant Frontend for Public Health Awareness

A near-instant dashboard offering total transparency. Features include:

- Observed current conditions
- 72-hour interactive predictions
- Historical trends & temperature seasonality
- Linear SHAP explainability for model transparency



A Blueprint for Unmonitored Regions

**Scaling Low-Cost
Satellite Forecasting
Globally**

karAQI proves that satellite-derived CAMS data paired with a 100% free CI/CD stack can accurately forecast AQI where physical infrastructure fails. With zero paid APIs or hosting, this architecture serves as a scalable, deployable blueprint for thousands of unmonitored municipalities worldwide.